

LearnPulse AI

AI-Powered Learning Management System with Contextual AI Tutor

WEEK 2 IMPLEMENTATION REPORT

Backend Infrastructure, PostgreSQL + pgvector Setup, Dependency Integration, Standard API Contracts, & OpenAPI/Swagger Verification

Project Name:	AI-Powered Learning Management System with Contextual AI Tutor
Document Title:	Week 2 Implementation Report (Backend Infrastructure)
Official Phase:	WEEK 2 IMPLEMENTATION (Git Branch: week-2-implementation)
Technology Stack:	Spring Boot 3.2.5, Java 21, Maven 3.9.6, PostgreSQL 16, pgvector 0.8.2
Document Version:	1.0.0-FINAL
Report Date:	August 8, 2026
Verification Status:	COMPLETED & VERIFIED (BUILD SUCCESS — 100% Tests Pass)
Prepared For:	Project Mentor & Evaluation Committee

- **Project Title:** AI-Powered Learning Management System with Contextual AI Tutor
- **Document Title:** Week 2 Implementation Report
- **Phase:** Week 2 Implementation — Backend Infrastructure & Configuration
- **Focus:** Backend Infrastructure, PostgreSQL + pgvector Setup, Dependency Integration, Standard API Contracts, & OpenAPI/Swagger Verification
- **Technology Stack:** Spring Boot 3.2.5 + Java 21 + Maven 3.9.6 + PostgreSQL 16 + pgvector 0.8.2
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1. COVER / DOCUMENT INFORMATION

This document serves as the official, mentor-facing **Week 2 Implementation Report** for the AI-Powered Learning Management System with Contextual AI Tutor. It provides a complete, transparent, and empirical record of all technical tasks executed, architecture established, dependencies integrated, database configurations initialized, and automated tests performed during Week 2.

DOCUMENT CONTROL METADATA	
Project Name	: AI-Powered Learning Management System with Contextual AI Tutor
Document Title	: Week 2 Implementation Report
Official Phase	: WEEK 2 IMPLEMENTATION (Git Branch: week-2-implementation)
Primary Objective	: Backend Infrastructure & System Configuration
Target Runtime	: OpenJDK 21 (build 21.0.12+7-LTS)
Framework	: Spring Boot 3.2.5
Build System	: Apache Maven 3.9.6
Database Engine	: PostgreSQL 16.8 (Database: learning_assistant_db)
Vector Extension	: pgvector 0.8.2
API Specification	: OpenAPI 3.0 via SpringDoc OpenAPI 2.5.0
Report Date	: August 8, 2026
Author / Team	: Technical Architecture & Systems Solution Engineering Team
Target Audience	: Academic Mentor, Technical Evaluator, & Engineering Leads

2. EXECUTIVE SUMMARY

The primary objective of **Week 2** was to initialize a robust, production-ready **Spring Boot 3.2.5** backend foundation targeting **Java 21**, configure **PostgreSQL 16** with the native **pgvector** extension, integrate required backend dependencies, establish standardized API response and error messaging models, and configure OpenAPI / Swagger UI documentation.

Week 2 was strictly focused on **backend infrastructure and configuration**. In accordance with the mentor's explicit specification, no domain business modules (authentication endpoints, student/teacher workspaces, course management, quizzes) or AI/RAG engine features (text chunking, vector embedding generation, vector similarity searches, LLM completions) were implemented.

Key Achievements Summary:

1. **Spring Boot Backend Initialization:** Successfully initialized and configured a Spring Boot 3.2.5 application on Java 21 using Maven.
2. **Standardized Package Layout:** Established the 10 base application packages (config, controller, service, repository, entity, dto, security, exception, util, ai).
3. **Dependency Integration:** Integrated all required dependencies into pom.xml (Spring Web, Spring Data JPA, Spring Security, Bean Validation, PostgreSQL Driver, Apache Tika, Apache PDFBox, Lombok, SpringDoc OpenAPI) without dependency conflicts.
4. **PostgreSQL & pgvector Configuration:** Connected to learning_assistant_db via HikariCP connection pooling and verified pgvector version 0.8.2 at both the database level (SQL queries) and application level (Spring Data JPA native

- repository queries).
5. **Standard API Response Wrapper:** Created `ApiResponse<T>` providing a unified JSON structure (status, message, data, timestamp, errors) across successful responses and error payloads.
6. **Global Exception Handling:** Built a `@RestControllerAdvice (GlobalExceptionHandler)` to convert uncaught runtime errors and Bean Validation failures into standardized error responses.
7. **OpenAPI / Swagger UI Setup:** Configured SpringDoc OpenAPI with project metadata and Bearer JWT security scheme documentation support. Interactive Swagger UI is accessible at `/swagger-ui.html`.
8. **Base Status API & Infrastructure Verification:** Implemented `GET /api/v1/status` and executed automated JUnit 5 / MockMvc tests, achieving ****100% build success (BUILD SUCCESS)**** and zero test failures.

3. WEEK 2 OBJECTIVES

The table below maps the mentor's explicit Week 2 specifications against the actual technical implementation and verification evidence:

Mentor Requirement	Implementation Status	Evidence / Verification
1. Initialize Spring Boot Backend	Completed	Spring Boot 3.2.5 application initialized on Java 21; main class <code>LearningAssistantApplication</code> compiles cleanly.
2. Base Package Structure	Completed	Created 10 required base packages: <code>config</code> , <code>controller</code> , <code>service</code> , <code>repository</code> , <code>entity</code> , <code>dto</code> , <code>security</code> , <code>exception</code> , <code>util</code> , <code>ai</code> .
3. Base Status/Health Endpoint	Completed	Implemented <code>BaseStatusController</code> at <code>GET /api/v1/status</code> returning environment health metrics.
4. Integrate Maven Dependencies	Completed	<code>pom.xml</code> configured with Spring Web, Data JPA, Security, Postgres, Validation, Tika, PDFBox, Lombok, SpringDoc.
5. Configure PostgreSQL Connection	Completed	<code>application.yml</code> configured for <code>learning_assistant_db</code> ; HikariCP pool initialized with 10 max connections.
6. Install & Verify pgvector	Completed	<code>pgvector</code> extension <code>v0.8.2</code> installed in PostgreSQL and verified programmatically via <code>InfrastructureCheckRepository</code> .
7. Standard API Response Structure	Completed	Implemented generic <code>ApiResponse<T></code> wrapper returning status, message, data, timestamp, and errors.
8. Global Error Handling Foundation	Completed	Created <code>GlobalExceptionHandler</code> with <code>@RestControllerAdvice</code> handling <code>ApiException</code> and validation errors.
9. Configure OpenAPI & Swagger UI	Completed	Configured <code>OpenApiConfig</code> with project metadata, Bearer auth documentation scheme, and Swagger UI at <code>/swagger-ui.html</code> .
10. Infrastructure Verification Tests	Completed	Executed <code>mvn clean test</code> ; 100% test pass rate (<code>LearningAssistantApplicationTests</code> and <code>BaseStatusControllerTest</code>).

4. SCOPE OF IMPLEMENTATION

The scope of Week 2 was strictly constrained to establishing backend infrastructure and system configuration.

In Scope for Week 2:

- Spring Boot project initialization on Java 21 with Maven.
- Implementation of the 10 required base package directories.
- Dependency configuration in `pom.xml` (Web, Data JPA, Security, Postgres, Bean Validation, Tika, PDFBox, Lombok, SpringDoc OpenAPI).
- PostgreSQL connection setup in `application.yml` for `learning_assistant_db`.
- HikariCP connection pooling and Hibernate JPA properties configuration.
- Installation and programmatic verification of `pgvector` in PostgreSQL.
- Implementation of generic `ApiResponse<T>` wrapper and `GlobalExceptionHandler`.
- Security filter chain foundation permitting public access to status and Swagger UI endpoints.
- OpenAPI metadata configuration and Swagger UI setup.
- Base status endpoint GET `/api/v1/status`.
- Infrastructure verification via unit and integration tests.

IMPORTANT:

Explicit Scope Boundary: Week 2 was limited exclusively to backend infrastructure and configuration. No business domain modules (User, Course, Quiz) or AI retrieval features (Embeddings, RAG, Vector Search, LLM) were implemented.

5. PROJECT STRUCTURE

The following tree represents the complete file structure implemented in the repository:

```

lms internship/
■■■ pom.xml # Maven project configuration & dependency management
■■■ application.yml # Centralized Spring Boot application configuration
■■■ WEEK_2_IMPLEMENTATION_REPORT.md # Official Week 2 Markdown Implementation Report
■■■ WEEK_2_IMPLEMENTATION_REPORT.pdf # Polished Mentor-Facing PDF Implementation Report
■■■ src/
    ■■■ main/
        ■■■ java/
            ■■■ com/learnpulse/backend/
            ■■■ LearningAssistantApplication.java # Spring Boot Main Entrypoint
            ■■■ ai/
            ■■■ AiMarker.java # AI Package Marker Interface (Foundation)
            ■■■ config/
            ■■■ OpenApiConfig.java # SpringDoc OpenAPI 3.0 & Swagger Config
            ■■■ controller/
            ■■■ BaseStatusController.java # REST Controller for GET /api/v1/status
            ■■■ dto/
            ■■■ ApiResponse.java # Standardized Generic API Response DTO
            ■■■ entity/
            ■■■ InfrastructureCheckEntity.java # Minimal Entity for JPA Schema Verification
            ■■■ exception/
            ■■■ ApiException.java # Custom Runtime API Exception Class
            ■■■ GlobalExceptionHandler.java # Standardized RestControllerAdvice Handler
            ■■■ repository/
            ■■■ InfrastructureCheckRepository.java # JPA Repo for pgvector & DB Verification
            ■■■ security/
            ■■■ SecurityConfig.java # Spring Security FilterChain Configuration
            ■■■ service/
            ■■■ BaseStatusService.java # Service Aggregating System Health Metrics
            ■■■ util/
            ■■■ DateUtil.java # ISO-8601 Date Formatting Utility Class
        ■■■ resources/
        ■■■ application.yml # Database, Server, & JPA Properties
    ■■■ test/
        ■■■ java/
            ■■■ com/learnpulse/backend/
            ■■■ LearningAssistantApplicationTests.java # Context Load Integration Test
            ■■■ controller/
            ■■■ BaseStatusControllerTest.java # Base Status MockMvc Integration Test

```

Class Responsibilities Summary:

- **LearningAssistantApplication:** Initializes Spring ApplicationContext and boots embedded Tomcat.
- **AiMarker:** Foundational marker interface in the ai package indicating where future RAG/AI components will reside.
- **OpenApiConfig:** Defines OpenAPI 3.0 document info, contact details, and Bearer JWT security scheme for Swagger UI.
- **BaseStatusController:** Exposes GET /api/v1/status returning ApiResponse<StatusData>.
- **ApiResponse:** Generic response wrapper encapsulating status, message, data, timestamp, and errors.
- **InfrastructureCheckEntity:** Lightweight JPA entity mapped to infrastructure_checks table to test Hibernate DDL auto-generation.
- **ApiException:** Runtime exception thrown for controlled infrastructure or application error conditions.
- **GlobalExceptionHandler:** Global exception advice intercepting exceptions and returning standardized HTTP 4xx/5xx ApiResponse payloads.
- **InfrastructureCheckRepository:** Spring Data JPA repository executing native PostgreSQL queries to check pgvector presence.
- **SecurityConfig:** Configures stateless HTTP security permitting public access to /api/v1/status, /swagger-ui/**, and /v3/api-docs/**.
- **BaseStatusService:** Service collecting environment data (Java version, DB connection status, pgvector version).
- **DateUtil:** Utility formatting Instant timestamps into ISO-8601 strings.

6. LAYERED ARCHITECTURE

The application adopts a clean, layered package architecture. The table below details the purpose, implementation status, and future role of each base package:

Package Name	Implemented Code	Purpose & Implemented Responsibility	Foundation Status
config	OpenApiConfig	Registers OpenAPI 3.0 beans and Swagger UI documentation settings.	Foundation for future API documentation.
controller	BaseStatusController	Handles HTTP REST requests and returns standardized <code>ResponseEntity<ApiResponse<T>></code> .	Foundation for future REST controllers.
service	BaseStatusService	Encapsulates system health aggregation and database status retrieval logic.	Foundation for future business service layer.
repository	InfrastructureCheckRepository	Executes Spring Data JPA and native PostgreSQL SQL verification queries.	Foundation for future database repositories.
entity	InfrastructureCheckEntity	Defines JPA entity mappings and Hibernate database schema generation.	Foundation for future domain entities.
dto	ApiResponse	Defines generic request/response data transfer contracts.	Foundation for future API request/response DTOs.
security	SecurityConfig	Defines Spring Security <code>SecurityFilterChain</code> permitting public endpoints.	Foundation for future JWT auth filters.
exception	GlobalExceptionHandler, ApiException	Provides centralized <code>@RestControllerAdvice</code> error handling and custom exceptions.	Foundation for future domain exception handling.
util	DateUtil	Provides static utility helper functions.	Foundation for future utility classes.
ai	AiMarker (Marker Interface)	Foundational package marker interface as explicitly required by the mentor.	Marker foundation for future AI/RAG modules.

NOTE:

Package Transparency: The `ai` package currently contains `AiMarker.java` as a foundational marker interface. No AI models, vector stores, or LLM code were implemented inside this package during Week 2.

7. SPRING BOOT INITIALIZATION

The Spring Boot backend was initialized with the following technical specifications:

- **Spring Boot Version:** 3.2.5
- **Java SDK Version:** Java 21 (OpenJDK 21.0.12 LTS)
- **Build Tool:** Apache Maven 3.9.6
- **Embedded Web Server:** Tomcat 10.1.20 (Default port: 8080)
- **Main Class:** `com.learnpulse.backend.LearningAssistantApplication`

Main Application Source Code:

```
package com.learnpulse.backend;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class LearningAssistantApplication {

    public static void main(String[] args) {
        SpringApplication.run(LearningAssistantApplication.class, args);
    }
}
```

Verification Evidence:

Application startup was verified via Spring Boot context loading tests (`LearningAssistantApplicationTests.java`) and command-line execution:

```

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=====|_|=====|_|=/ / / / /
:: Spring Boot ::                (v3.2.5)

```

```
2026-08-08T19:58:53.762+05:30 INFO 51776 --- [learning-assistant] [main] c.l.b.LearningAssistantApplicationTests :
Starting LearningAssistantApplicationTests using Java 21.0.12 with PID 51776
2026-08-08T19:58:54.272+05:30 INFO 51776 --- [learning-assistant] [main] c.l.b.LearningAssistantApplicationTests :
Started LearningAssistantApplicationTests in 0.527 seconds
```

8. MAVEN DEPENDENCIES

The table below lists every dependency present in the project's `pom.xml`, including version, purpose, and current vs future usage:

Group ID & Artifact ID	Version	Primary Purpose	Week 2 Usage	Future / Current
org.springframework.boot:spring-boot-starter-web	3.2.5	Spring MVC REST Framework	Exposes REST APIs & embedded Tomcat server	Current & Future
org.springframework.boot:spring-boot-starter-data-jpa	3.2.5	ORM & Database Persistence	Manages PostgreSQL HikariCP pool & repositories	Current & Future
org.springframework.boot:spring-boot-starter-security	3.2.5	Application Security	Configures HTTP SecurityFilterChain foundation	Current & Future
org.springframework.boot:spring-boot-starter-validation	3.2.5	Bean Validation Framework	Validates request payloads & error handling	Current & Future
org.postgresql:postgresql	Runtime	PostgreSQL JDBC Driver	Connects backend to learning_assistant_db	Current & Future
org.projectlombok:lombok	Optional	Boilerplate Code Reduction	Generates getters, setters, builders, constructors	Current & Future
org.apache.tika:tika-core	2.9.2	Document Text Parsing	Dependency registered (No document parsing code written)	Future Week Scope
org.apache.tika:tika-parsers-standard-package	2.9.2	Standard Text Parsers	Dependency registered (No document parsing code written)	Future Week Scope

org.apache.pdfbox:pdfbox	3.0.2	PDF Processing Library	Dependency registered (No PDF processing code written)	Future Week Scope
org.springdoc:springdoc-openapi-starter-webmvc-ui	2.5.0	OpenAPI 3.0 & Swagger UI	Auto-generates OpenAPI docs & Swagger UI	Current & Future
org.springframework.boot:spring-boot-starter-test	Test	JUnit 5 & MockMvc Testing	Runs unit and integration test suite	Current & Future
org.springframework.security:spring-security-test	Test	Security Mocking Utilities	Used for testing secure security filter chains	Current & Future

NOTE:

Dependency Transparency: Apache Tika, Apache PDFBox, and Spring Security were included in pom.xml as required by the mentor for foundational environment readiness. Including these dependencies did NOT involve writing document ingestion or user authentication business logic in Week 2.

9. POSTGRESQL CONFIGURATION

The Spring Boot application connects to PostgreSQL 16 hosting the database `learning_assistant_db`.

- **Database Name:** `learning_assistant_db`
- **PostgreSQL Engine Version:** PostgreSQL 16.8 (Ubuntu 16.8-1.pgdg22.04+1)
- **JDBC Driver:** `org.postgresql.Driver`
- **HikariCP Connection Pool:** Name: `LearningAssistantHikariCP`, Max Pool Size: 10, Minimum Idle: 5, Idle Timeout: 300000ms, Connection Timeout: 20000ms.
- **JPA / Hibernate Dialect:** `org.hibernate.dialect.PostgreSQLDialect`
- **Schema Auto-Generation:** `hibernate.ddl-auto: update`

Connection Security:

Database credentials are managed safely using Spring environment property placeholders (`${DB_HOST:localhost}`, `${DB_PORT:5432}`, `${DB_NAME:learning_assistant_db}`, `${DB_USERNAME:postgres}`, `${DB_PASSWORD:postgres}`), enabling environment variable overrides during production deployment without hardcoding secrets.

10. APPLICATION.YML CONFIGURATION

The complete `src/main/resources/application.yml` file is structured into clear functional groups:

```
server:
  port: 8080
  servlet:
    context-path: /

spring:
  application:
    name: learning-assistant

datasource:
  url: jdbc:postgresql://${DB_HOST:localhost}:${DB_PORT:5432}/${DB_NAME:learning_assistant_db}
  username: ${DB_USERNAME:postgres}
  password: ${DB_PASSWORD:postgres}
  driver-class-name: org.postgresql.Driver
  hikari:
    maximum-pool-size: 10
    minimum-idle: 5
    idle-timeout: 300000
    connection-timeout: 20000
    pool-name: LearningAssistantHikariCP

jpa:
  database-platform: org.hibernate.dialect.PostgreSQLDialect
  hibernate:
    ddl-auto: update
  show-sql: false
  properties:
    hibernate:
      format_sql: true

springdoc:
  api-docs:
    path: /v3/api-docs
  swagger-ui:
    path: /swagger-ui.html
    operationsSorter: alpha
    tagsSorter: alpha
```

Configuration Group Breakdown:

1. ****server****: Sets application HTTP port to 8080 with root context path /.
2. ****spring.datasource****: Configures PostgreSQL JDBC connection parameters and HikariCP connection pool settings.
3. ****spring.jpa****: Configures Hibernate PostgreSQL dialect, SQL formatting, and schema auto-update (ddl-auto: update).
4. ****springdoc****: Configures OpenAPI JSON endpoint at /v3/api-docs and Swagger UI at /swagger-ui.html.

11. PGVECTOR INSTALLATION AND VERIFICATION

The PostgreSQL **pgvector** extension was installed and verified in `learning_assistant_db`.

11.1 Database-Level Verification

The extension was enabled in the PostgreSQL instance using SQL:

```
CREATE EXTENSION IF NOT EXISTS vector;
```

Execution of the verification query confirmed pgvector installation:

```
SELECT extname, extversion FROM pg_extension WHERE extname = 'vector';
```

Database Result:

```
extname | extversion  
-----+-----  
vector  | 0.8.2  
(1 row)
```

11.2 Application-Level Verification

The Spring Boot backend programmatically verifies pgvector availability upon calling `BaseStatusService`. It executes a native SQL query through `InfrastructureCheckRepository`:

```
public interface InfrastructureCheckRepository extends JpaRepository<InfrastructureCheckEntity, Long> {  
  
    @Query(value = "SELECT COUNT(*) > 0 FROM pg_extension WHERE extname = 'vector'", nativeQuery = true)  
    boolean isPgVectorInstalled();  
  
    @Query(value = "SELECT extversion FROM pg_extension WHERE extname = 'vector'", nativeQuery = true)  
    String getPgVectorVersion();  
}
```

When GET `/api/v1/status` is called, the JSON response confirms:

```
"database": {  
  "databaseName": "learning_assistant_db",  
  "connected": true,  
  "pgvectorInstalled": true,  
  "pgvectorVersion": "0.8.2"  
}
```

IMPORTANT:

Verification vs Feature Distinction: Installing and programmatically verifying pgvector confirms database readiness. Vector embeddings, passage chunking, cosine similarity queries, and RAG pipelines were NOT implemented during Week 2.

12. STANDARD API RESPONSE

All API responses follow a unified generic wrapper format (`ApiResponse<T>`).

Java DTO Implementation:

```

package com.learnpulse.backend.dto;

import lombok.AllArgsConstructor;
import lombok.Builder;
import lombok.Data;
import lombok.NoArgsConstructor;
import lombok.NoArgsConstructor;

import java.time.Instant;
import java.util.List;

@Data
@Builder
@NoArgsConstructor
@AllArgsConstructor
public class ApiResponse<T> {

    private String status;
    private String message;
    private T data;
    private String timestamp;
    private List<String> errors;

    public static <T> ApiResponse<T> success(String message, T data) {
        return ApiResponse.<T>builder()
            .status("success")
            .message(message)
            .data(data)
            .timestamp(Instant.now().toString())
            .errors(null)
            .build();
    }

    public static <T> ApiResponse<T> error(String message, List<String> errors) {
        return ApiResponse.<T>builder()
            .status("error")
            .message(message)
            .data(null)
            .timestamp(Instant.now().toString())
            .errors(errors)
            .build();
    }
}

```

Successful Response JSON Example (GET /api/v1/status):

```

{
  "status": "success",
  "message": "Backend infrastructure operating successfully",
  "data": {
    "application": "learning-assistant",
    "environmentStatus": "UP",
    "javaVersion": "21.0.12",
    "springBootVersion": "3.2.5",
    "database": {
      "databaseName": "learning_assistant_db",
      "connected": true,
      "pgvectorInstalled": true,
      "pgvectorVersion": "0.8.2"
    },
    "infrastructureChecks": {
      "jpaAutoSchema": "enabled",
      "hikariConnectionPool": "active",
      "beanValidation": "active",
      "springSecurityFoundation": "active",
      "openApiSwaggerSupport": "active",
      "documentDependencies": "Apache Tika & PDFBox registered"
    }
  },
  "timestamp": "2026-08-08T19:58:54.272Z",
  "errors": null
}

```

Error Response JSON Example (Validation / Application Failure):

```
{
  "status": "error",
  "message": "Validation failed",
  "data": null,
  "timestamp": "2026-08-08T19:58:54.300Z",
  "errors": [
    "Field 'email' must be a valid email address",
    "Field 'password' cannot be blank"
  ]
}
```

13. ERROR HANDLING

Centralized error handling is implemented via `GlobalExceptionHandler` using `@RestControllerAdvice`:

```
package com.learnpulse.backend.exception;

import com.learnpulse.backend.dto.ApiResponse;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.validation.FieldError;
import org.springframework.web.bind.MethodArgumentNotValidException;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.RestControllerAdvice;

import java.util.List;
import java.util.stream.Collectors;

@RestControllerAdvice
public class GlobalExceptionHandler {

    @ExceptionHandler(ApiException.class)
    public ResponseEntity<ApiResponse<Void>> handleApiException(ApiException ex) {
        ApiResponse<Void> response = ApiResponse.error(ex.getMessage(), List.of(ex.getMessage()));
        return new ResponseEntity<>(response, ex.getStatus());
    }

    @ExceptionHandler(MethodArgumentNotValidException.class)
    public ResponseEntity<ApiResponse<Void>> handleValidationExceptions(MethodArgumentNotValidException ex) {
        List<String> errors = ex.getBindingResult()
            .getFieldErrors()
            .stream()
            .map(FieldError::getDefaultMessage)
            .collect(Collectors.toList());

        ApiResponse<Void> response = ApiResponse.error("Validation failed", errors);
        return new ResponseEntity<>(response, HttpStatus.BAD_REQUEST);
    }

    @ExceptionHandler(Exception.class)
    public ResponseEntity<ApiResponse<Void>> handleGeneralException(Exception ex) {
        ApiResponse<Void> response = ApiResponse.error(
            "An unexpected server error occurred: " + ex.getMessage(),
            List.of(ex.getClass().getSimpleName())
        );
        return new ResponseEntity<>(response, HttpStatus.INTERNAL_SERVER_ERROR);
    }
}
```

Errors Currently Handled:

1. **ApiException**: Custom infrastructure/application exceptions with dynamic HTTP status codes.
2. **MethodArgumentNotValidException**: Spring Bean Validation errors (extracts field validation error messages).

3. **Exception**: Catch-all for uncaught system exceptions, preventing unformatted 500 error stack traces from leaking to clients.

14. BASE STATUS API

The backend exposes a foundational infrastructure status API:

- **HTTP Method**: GET
- **Path**: /api/v1/status
- **Access Level**: Public (PermitAll)
- **Controller Class**: BaseStatusController
- **Service Class**: BaseStatusService

Technical Checks Executed by Status API:

1. Verifies Spring ApplicationContext startup and system property retrieval (java.version, springBootVersion).
2. Tests active connection to PostgreSQL 16 database (learning_assistant_db).
3. Queries PostgreSQL catalog for pgvector extension presence and returns extension version (0.8.2).
4. Verifies active status of HikariCP pool, JPA auto-schema generation, Bean Validation, Spring Security, and OpenAPI.

15. OPENAPI / SWAGGER

SpringDoc OpenAPI 2.5.0 is configured in OpenApiConfig.java:

```
package com.learnpulse.backend.config;

import io.swagger.v3.oas.models.Components;
import io.swagger.v3.oas.models.OpenAPI;
import io.swagger.v3.oas.models.info.Contact;
import io.swagger.v3.oas.models.info.Info;
import io.swagger.v3.oas.models.security.SecurityRequirement;
import io.swagger.v3.oas.models.security.SecurityScheme;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;

@Configuration
public class OpenApiConfig {

    private static final String SECURITY_SCHEME_NAME = "Bearer Authentication";

    @Bean
    public OpenAPI customOpenAPI() {
        return new OpenAPI()
            .info(new Info()
                .title("LearnPulse AI - Learning Assistant Backend API")
                .description("Backend Infrastructure & Foundation APIs for LearnPulse AI Learning Management System")
                .version("1.0.0")
                .contact(new Contact()
                    .name("LearnPulse Engineering Team")
                    .email("engineering@learnpulse.ai")))
            .addSecurityItem(new SecurityRequirement().addList(SECURITY_SCHEME_NAME))
            .components(new Components()
                .addSecuritySchemes(SECURITY_SCHEME_NAME, new SecurityScheme()
                    .name(SECURITY_SCHEME_NAME)
                    .type(SecurityScheme.Type.HTTP)
                    .scheme("bearer")
                    .bearerFormat("JWT")
                    .description("Standard Authorization header using Bearer scheme.")));
    }
}
```

- **Swagger UI URL**: <http://localhost:8080/swagger-ui.html>

- **OpenAPI Spec URL:** `http://localhost:8080/v3/api-docs`
- **Currently Documented Endpoints:** `GET /api/v1/status`

NOTE:

Authentication Header Documentation vs Implementation: Configuring the Bearer Authentication security scheme in `OpenApiConfig` enables Swagger UI to render an "Authorize" button so future secured endpoints can document an Authorization: Bearer <token> header. Authentication logic (login, JWT parsing, token generation) was NOT implemented in Week 2.

16. SECURITY FOUNDATION

Spring Security is configured in `SecurityConfig.java` to establish a stateless HTTP security foundation:

```
package com.learnpulse.backend.security;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.security.config.annotation.web.builders.HttpSecurity;
import org.springframework.security.config.annotation.web.configuration.EnableWebSecurity;
import org.springframework.security.config.annotation.web.configurers.AbstractHttpConfigurer;
import org.springframework.security.config.http.SessionCreationPolicy;
import org.springframework.security.web.SecurityFilterChain;

@Configuration
@EnableWebSecurity
public class SecurityConfig {

    @Bean
    public SecurityFilterChain securityFilterChain(HttpSecurity http) throws Exception {
        http
            .csrf(AbstractHttpConfigurer::disable)
            .sessionManagement(session -> session.sessionCreationPolicy(SessionCreationPolicy.STATELESS))
            .authorizeHttpRequests(auth -> auth
                .requestMatchers(
                    "/api/v1/status/**",
                    "/v3/api-docs/**",
                    "/swagger-ui/**",
                    "/swagger-ui.html"
                ).permitAll()
                .anyRequest().authenticated()
            );

        return http.build();
    }
}
```

Current Security Policy:

- Disables CSRF for stateless REST API processing.
- Configures STATELESS session management (no HTTP session creation).
- Explicitly permits public access to `/api/v1/status/**`, `/v3/api-docs/**`, `/swagger-ui/**`, and `/swagger-ui.html`.
- Mandates authentication for all unlisted endpoints (`anyRequest().authenticated()`).

WARNING:

Authentication Disclosure: Authentication and authorization mechanisms (user credentials, password encoding, JWT token generation, role checks) are **NOT implemented in Week 2**.

17. TESTING AND VERIFICATION

Automated testing was performed using JUnit 5, Spring Boot Test, and MockMvc.

17.1 Test Classes Executed

1. LearningAssistantApplicationTests.java

- **Test Type:** Integration Test (@SpringBootTest)
- **Purpose:** Verifies that the Spring ApplicationContext loads cleanly on Java 21, initializing HikariCP, JPA EntityManagerFactory, and SecurityFilterChain without bean creation errors.
- **Result:** ****PASS**** (Execution time: 3.474s)

2. BaseStatusControllerTest.java

- **Test Type:** Controller MockMvc Integration Test (@SpringBootTest, @AutoConfigureMockMvc)
- **Purpose:** Verifies that GET /api/v1/status returns HTTP 200 OK with the exact ApiResponse<StatusData> JSON contract structure, validating status: "success", application name, database connection state, pgvectorInstalled: true, and errors: null.
- **Result:** ****PASS**** (Execution time: 0.707s)

17.2 Maven Test Suite Output

Execution command:

```
.tools/apache-maven-3.9.6/bin/mvn clean test
```

Build & Test Log Output:

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.learnpulse.backend.LearningAssistantApplicationTests
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.474 s -- in com.learnpulse.backend.Learning
AssistantApplicationTests
[INFO] Running com.learnpulse.backend.controller.BaseStatusControllerTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.707 s -- in com.learnpulse.backend.controll
er.BaseStatusControllerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 7.738 s
[INFO] Finished at: 2026-08-08T19:58:54+05:30
[INFO] -----
```

18. FILES CREATED AND MODIFIED

The table below documents every file created or modified during Week 2:

File Path	Change Type	Technical Purpose
pom.xml	Modified / Created	Maven project descriptor defining dependencies, Java 21 properties, and build plugins
src/main/resources/application.yml	Created	Application configuration settings for Server, Datasource, HikariCP, JPA, and SpringDoc

LearningAssistantApplication.java	Created	Main Spring Boot application entrypoint class
ApiResponse.java	Created	Generic API response wrapper DTO (status, message, data, timestamp, errors)
OpenApiConfig.java	Created	OpenAPI 3.0 bean configuration and Bearer authorization security scheme
SecurityConfig.java	Created	Spring Security SecurityFilterChain permitting status and Swagger UI routes
BaseStatusController.java	Created	REST controller handling GET /api/v1/status
BaseStatusService.java	Created	Service class gathering system metrics, DB status, and pgvector version
InfrastructureCheckRepository.java	Created	Spring Data JPA repository executing native PostgreSQL SQL queries for pgvector
InfrastructureCheckEntity.java	Created	JPA entity checking table creation and schema auto-generation
GlobalExceptionHandler.java	Created	Centralized @RestControllerAdvice error handler
ApiException.java	Created	Custom runtime exception class for controlled application error handling
DateUtil.java	Created	ISO-8601 timestamp formatting utility
AIMarker.java	Created	Package marker interface establishing foundational ai package
LearningAssistantApplicationTests.java	Created	Context loading integration test
BaseStatusControllerTest.java	Created	MockMvc controller integration test for /api/v1/status
WEEK_2_IMPLEMENTATION_REPORT.md	Created	Official Week 2 Markdown Implementation Report
WEEK_2_IMPLEMENTATION_REPORT.pdf	Created	Official Week 2 PDF Implementation Report

19. REQUIREMENT-BY-REQUIREMENT CHECKLIST

Below is the day-by-day mentor requirement compliance audit:

Mentor Requirement	Target Day	Status	Empirical Evidence
Initialize Spring Boot backend on Java 21 & Maven	Day 7	Completed	pom.xml configured for Java 21; LearningAssistantApplication compiles cleanly.
Create 10 base packages (config, controller, service, repository, entity, dto, security, exception, util, ai)	Day 7	Completed	All 10 package directories created under com.learnpulse.backend.
Create base status/health endpoint (GET /api/v1/status)	Day 7	Completed	BaseStatusController exposes /api/v1/status returning environment status metrics.
Verify application startup and basic status API	Day 7	Completed	MockMvc test BaseStatusControllerTest passes cleanly.

Add required dependencies (Web, JPA, Security, Postgres, Validation, Tika, PDFBox, Lombok, SpringDoc)	Day 8	Completed	Dependencies verified in <code>pom.xml</code> ; project compiles with zero conflicts.
Configure PostgreSQL database connection (<code>learning_assistant_db</code>)	Day 9	Completed	<code>application.yml</code> configured; HikariCP connection pool successfully established.
Configure HikariCP connection pooling & JPA auto-schema generation	Day 9	Completed	HikariCP pool initialized (max 10, min idle 5); <code>hibernate.ddl-auto: update</code> active.
Install & verify PostgreSQL <code>pgvector</code> extension	Day 9	Completed	<code>pgvector v0.8.2</code> installed in PostgreSQL and verified programmatically via SQL query.
Implement generic API response wrapper (<code>ApiResponse<T></code>)	Day 10	Completed	<code>ApiResponse</code> DTO contains status, message, data, timestamp, errors.
Implement global exception handler foundation	Day 10	Completed	<code>GlobalExceptionHandler @RestControllerAdvice</code> returns standardized <code>ApiResponse</code> on error.
Configure SpringDoc OpenAPI & Swagger UI (<code>/swagger-ui.html</code>)	Day 11	Completed	<code>OpenApiConfig</code> configures OpenAPI 3.0 spec; Swagger UI accessible at <code>/swagger-ui.html</code> .
Configure Bearer authentication header documentation support	Day 11	Completed	<code>OpenApiConfig</code> registers HTTP Bearer security scheme in OpenAPI components.
Verify Spring Security filter chain foundation	Day 11	Completed	<code>SecurityConfig</code> permits <code>/api/v1/status</code> , <code>/v3/api-docs/**</code> , <code>/swagger-ui/**</code> .
Perform complete infrastructure verification test suite	Day 12	Completed	<code>mvn clean test</code> succeeds with <code>**BUILD SUCCESS**</code> (2/2 tests pass).

20. ITEMS INTENTIONALLY NOT IMPLEMENTED

To maintain total transparency, the following modules were **deliberately left unimplemented** during Week 2 because they fall outside the Week 2 scope:

- User Authentication & Authorization:** No user login endpoints (`/login`), user registration (`/register`), JWT token generation, password hashing (`BCryptPasswordEncoder`), or `SecurityContext` user role authorization.
- User Management Domain:** No User, Student, or Teacher domain entity tables, repositories, services, or REST controllers.
- Course & Syllabus Management:** No Subject, Chapter, Note, or PDF Document management models, repositories, file upload services, or endpoints.
- Assessment & Grading Engine:** No Quiz, Question, Option, `QuizAttempt`, or automated scoring engines.
- AI Subsystem & RAG Pipeline:** No text passage chunking, vector embedding generation, `pgvector` vector similarity queries, system prompt assembly, LLM integration, or SSE streaming endpoints.
- Full LMS Database Tables:** Full relational database schema tables designed in Week 1 remain unimplemented until their designated implementation phases.

21. SCOPE / DEVIATION DISCLOSURE

Disclosure Statement:

To enable programmatic verification of JPA auto-schema generation and native SQL execution for pgvector detection, two lightweight infrastructure helper classes were created:

1. ****InfrastructureCheckEntity****: A minimal JPA entity mapped to `infrastructure_checks` to test Hibernate table creation.
2. ****InfrastructureCheckRepository****: A Spring Data JPA repository executing native PostgreSQL queries (`SELECT COUNT(*) > 0 FROM pg_extension WHERE extname = 'vector'`).

Deviation Impact Assessment:

- **Why Added**: Created strictly to programmatically test PostgreSQL database connectivity and pgvector extension availability during the base status check (`GET /api/v1/status`).
- **Functionality Impact**: Zero negative impact. Does not implement business logic or domain entities.
- **Recommendation**: Retain in repository as an infrastructure health verification mechanism.

NOTE:

*Other than the two infrastructure verification helper classes listed above, **no material deviations from the mentor's Week 2 specification were identified.***

22. KNOWN LIMITATIONS

The current implementation has the following known technical boundaries:

1. **No Authentication Layer**: Endpoints other than public status and Swagger UI routes require authentication, but no login or token generation logic exists yet.
2. **Local Environment Database Configuration**: Datasource configuration defaults to `localhost:5432` with standard development credentials (`postgres/postgres`). Production deployment will require environment variable overrides (`DB_HOST`, `DB_PASSWORD`).
3. **No Domain Business Logic**: The backend currently serves infrastructure status queries only.
4. **Document Dependencies Standby**: Apache Tika and PDFBox libraries are present in `pom.xml`, but no document ingestion service classes have been created yet.

23. WEEK 2 COMPLETION SUMMARY

Week 2 has achieved all objectives established by the mentor. A stable, high-performance Spring Boot 3.2.5 backend foundation targeting Java 21 has been initialized and verified. PostgreSQL 16 connectivity and pgvector v0.8.2 installation have been programmatically confirmed. Standardized API messaging contracts, global error handling, Spring Security filter chain foundation, and interactive Swagger UI documentation are active and tested.

All verification steps passed with ****BUILD SUCCESS and 100% automated test pass rate****. Week 2 is ready to close.

24. READINESS FOR WEEK 3

The established backend infrastructure provides a verified, production-ready foundation for future development phases. The project is completely prepared for the next week of development to build domain entities, database repositories, user authentication, and business REST APIs on top of this stable architecture.

End of Official Week 2 Implementation Report.